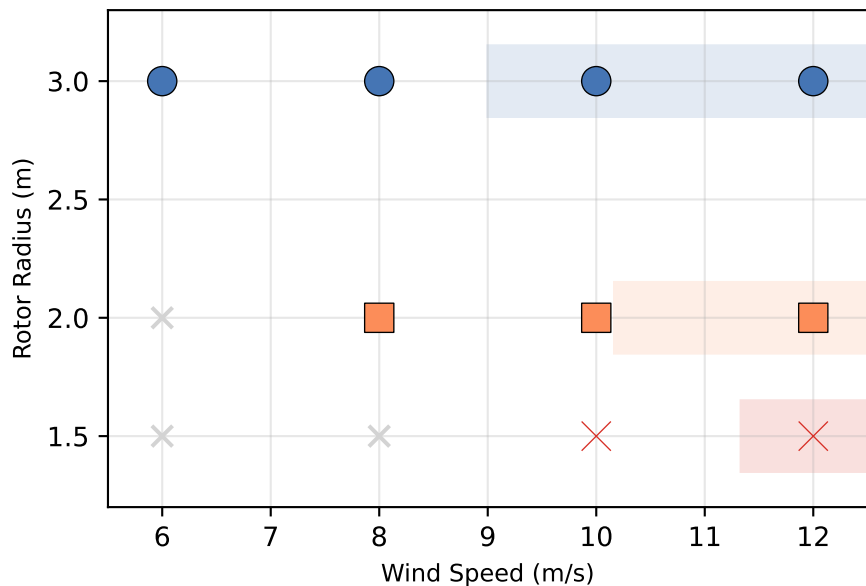
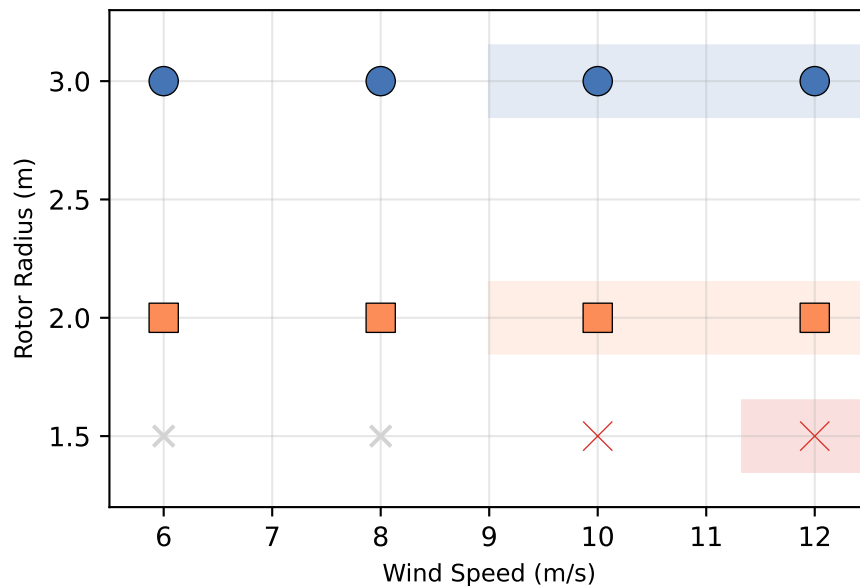


Operating Wind Envelope: Which (R,N) Combinations Fly? BEM v2.1 — Filled = Viable, X = No Tension or Overspeed

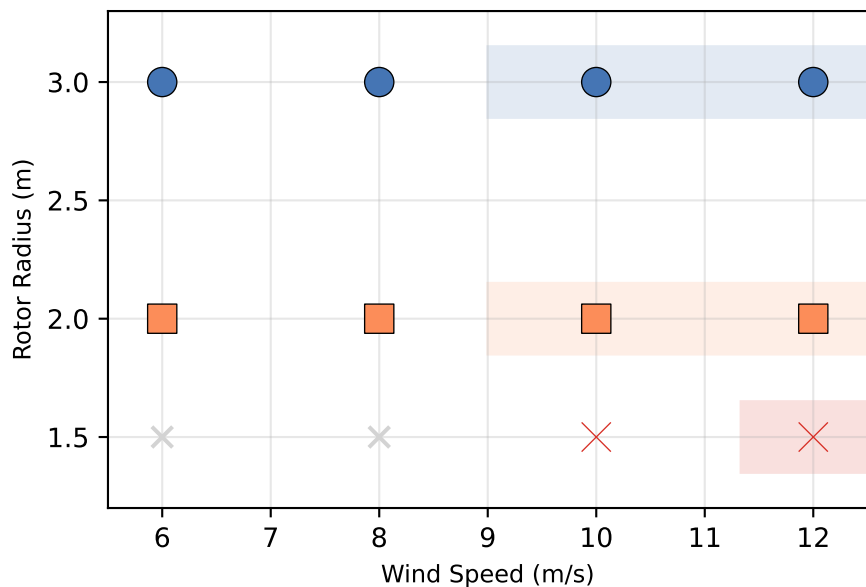
N = 1



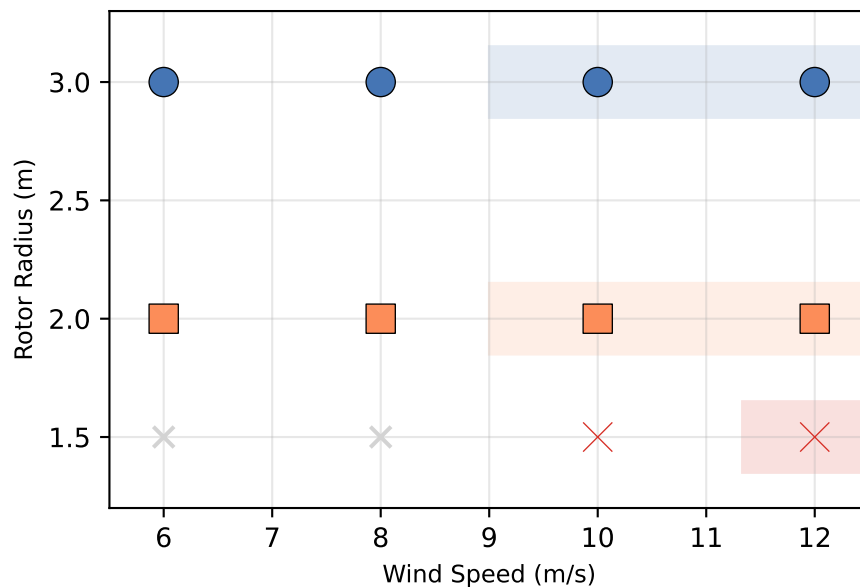
N = 2



N = 3



N = 4



IMPLICATIONS: R=1.5m produces zero tension at all N, all wind speeds — below the viability threshold for NACA 0012 at $Re < 5 \times 10^4$. R=2.0m flies at $N \geq 2$ above 8 m/s. R=3.0m flies across all conditions. Tip speeds stay under 40 m/s everywhere, well below the 120 m/s noise limit. No configurations exceed viability constraints. Data: BEM v2.1, 45° elevation.